SAT MATH

Practice tests • Practice Test 1 • Module 2, easier (22 questions, 35 minutes)

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Name: _____ Date: _____

Directions

These questions follow the format of the SAT Math section. A calculator is allowed on every question (Desmos is built into the digital SAT). Multiple-choice questions have one best answer. For **student-produced responses**, write your answer on the line: a fraction or a decimal is accepted (for example $3/4$ or 0.75); give negative answers with a minus sign; do not use mixed numbers.

Question 1 (Algebra)

If $x + 9 = 15$, what is the value of x?

(A) 24
(B) 6
(C) -6
(D) 9

Question 2 (Algebra)

If $4x = 28$, what is the value of x?

(A) 7
(B) 24
(C) 32
(D) 112

Question 3 (Data Analysis)

Three notebooks cost $7.50. At the same price, how much do five notebooks cost?

(A) $10.00
(B) $15.00
(C) $12.50
(D) $37.50

Question 4 (Advanced Math)

Which expression is equivalent to $3(x^2 + 2x) - x^2$?

(A) $2x^2 + 2x$
(B) $3x^2 + 6x$
(C) $4x^2 + 6x$
(D) $2x^2 + 6x$

Question 5 (Algebra) STUDENT-PRODUCED RESPONSE

The function f is defined by $f(x) = 2x + 7$. What is the value of $f(5)$?

Your answer: _____

Question 6 (Geometry and Trig)

A square has a perimeter of 36. What is its area?

(A) 36
(B) 81
(C) 144
(D) 9

Question 7 (Data Analysis)

What is the median of 2, 9, 4, 7, 5?

(A) 4
(B) 5.4
(C) 7
(D) 5

Question 8 (Algebra)

What is the slope of the line $y = -3x + 4$?

(A) -3
(B) 4
(C) 3
(D) $\frac{4}{3}$

Question 9 (Advanced Math)

If $x^2 = 36$ and $x > 0$, what is the value of x?

(A) 18
(B) 72
(C) 6
(D) -6

Question 10 (Algebra) STUDENT-PRODUCED RESPONSE

What is the solution to $2x + 3 = x + 10$?

Your answer: _____

Question 11 (Geometry and Trig)

In a right triangle, one acute angle measures $35°$. What is the measure of the other acute angle?

(A) $145°$
(B) $55°$
(C) $65°$
(D) $45°$

Question 12 (Data Analysis)

What is 25% of 80?

(A) 25
(B) 32
(C) 55
(D) 20

Question 13 (Advanced Math)

The function f is defined by $f(x) = x^2 + 1$. What is the value of $f(3)$?

(A) 10
(B) 7
(C) 9
(D) 16

Question 14 (Algebra)

A phone plan costs \$15 per month plus \$0.10 per text message. What is the cost of a month with 120 text messages?

(A) \$12
(B) \$135
(C) \$27
(D) \$16.20

Question 15 (Advanced Math) STUDENT-PRODUCED RESPONSE

What is the positive solution of $(x - 2)(x + 5) = 0$?

Your answer: _____

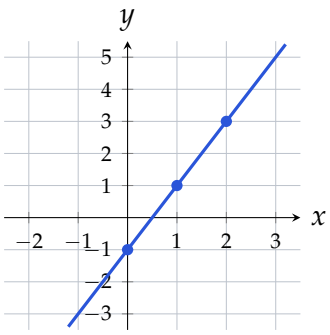
Question 16 (Geometry and Trig) STUDENT-PRODUCED RESPONSE

A circle has diameter 10. Its circumference is $k\pi$. What is the value of k?

Your answer: _____

Question 17 (Algebra)

Which equation represents the line shown?



(A) $y = x - 1$
(B) $y = -2x - 1$
(C) $y = 2x + 1$
(D) $y = 2x - 1$

Question 18 (Data Analysis)

A bag holds 3 red, 5 blue and 2 green marbles. One marble is drawn at random. What is the probability that it is blue?

(A) $\frac{1}{2}$
(B) $\frac{1}{5}$
(C) $\frac{5}{3}$
(D) $\frac{3}{10}$

Question 19 (Advanced Math)

Which expression is equivalent to $x^3 \cdot x^4$?

(A) x^{12}
(B) x^7
(C) $2x^7$
(D) x^1

Question 20 (Algebra) STUDENT-PRODUCED RESPONSE

If $x + y = 12$ and $x - y = 2$, what is the value of y?

Your answer: _____

Question 21 (Geometry and Trig)

The legs of a right triangle are 6 and 8. What is the length of the hypotenuse?

(A) 14
(B) $\sqrt{28}$
(C) 10
(D) 48

Question 22 (Advanced Math) STUDENT-PRODUCED RESPONSE

If $2^x = 64$, what is the value of x?

Your answer: _____